

Exact Rigid-Link Closure and Center-Distance Correction for Open Roller Chain Drives

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Abstract

This article presents a discrete geometric method for positioning the roller centers of an open two-sprocket chain drive and correcting its center distance. The analytical curve formed by the sprocket pitch-circle arcs and their common external tangents is retained as the roller-center locus and as a link-count estimate. Physical closure, however, is imposed by walking rigid chords of length equal to the chain pitch. For each candidate integer link count, the center distance is solved from the residual between the path coordinate reached after the last rigid pitch step and one complete traversal of the locus. The method reports link parity, offset-link requirement, closure residual and maximum pitch error, and returns link poses suitable for CAD generation. An ASA 80 example with 11- and 20-tooth sprockets is used as a regression case.

Keywords: roller chain; rigid-link closure; center distance; pitch chord; offset link; computational geometry; CAD.

1 Problem definition

Consider an open, coplanar, single-strand roller chain drive with parallel shafts. The input is the chain pitch p , sprocket tooth counts $z_1 \leq z_2$, and desired center distance a_0 . The required output is an integer number of links N , a corrected center distance a_f , and roller centers P_i such that

$$\|P_{i+1} - P_i\| = p, \quad i = 0, \dots, N-1, \quad P_N = P_0. \quad (1)$$

An odd value of N is geometrically admissible but requires an offset link in the alternating inner/outer assembly. An even-only option is therefore part of the design input rather than a post-processing rule.

2 Pitch geometry and continuous locus

The pitch radius of a sprocket with z teeth is

$$r = \frac{p}{2 \sin(\pi/z)}, \quad D_p = \frac{p}{\sin(\pi/z)}. \quad (2)$$

Let $\Delta r = r_2 - r_1$. The external tangents exist in the domain

$$a > |\Delta r|. \quad (3)$$

The tangent inclination and straight-span length are

$$\alpha = \arcsin\left(\frac{\Delta r}{a}\right), \quad j = \sqrt{a^2 - \Delta r^2}. \quad (4)$$

The wrap angles are

$$\theta_1 = \pi - 2\alpha, \quad \theta_2 = \pi + 2\alpha, \quad (5)$$

and the length of the complete continuous locus is

$$C_i(a) = r_1\theta_1 + r_2\theta_2 + 2j = \pi(r_1 + r_2) + 2\Delta r \arcsin\left(\frac{\Delta r}{a}\right) + 2\sqrt{a^2 - \Delta r^2}. \quad (6)$$

Equation (6) describes the roller-center locus. It is also an efficient initial estimator,

$$N_0 = \text{round}\left(\frac{C_i(a_0)}{p}\right). \quad (7)$$

It is *not* the final physical closure equation. On a sprocket the chain pitch is a chord between adjacent roller centers, whereas an arc contributes a larger arc length. Consequently, solving $C_i(a) = Np$ produces a center distance for a continuously flexible belt, not for the rigid-link polygon required by the CAD assembly.

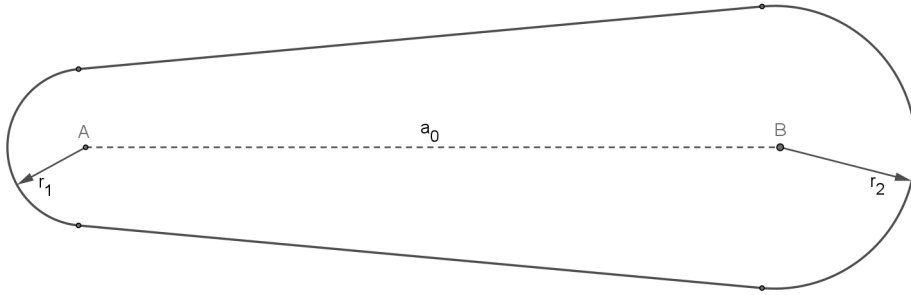


Figure 1: Open-drive pitch-circle geometry and common external tangents.

3 Discrete roller placement

Let $q(s; a)$ be a periodic point function on the continuous locus, parameterized by forward arc length $s \in [0, C_i(a))$. The locus consists, in order, of the small-sprocket arc, lower tangent, large-sprocket arc and upper tangent.

Starting from $s_0 = 0$, the next roller is the first forward point satisfying the chord condition:

$$s_{i+1} = \min_{s > s_i} \{s : \|q(s; a) - q(s_i; a)\| = p\}. \quad (8)$$

The scalar root in Equation (8) is bracketed by a forward scan and refined by bisection. This construction naturally crosses tangent–arc boundaries while preserving the same rigid pitch distance.

After N steps, define the discrete closure residual

$$g(a; N) = s_N(a) - C_i(a). \quad (9)$$

The corrected center distance is the physically valid root

$$g(a_f; N) = 0, \quad a_f > |\Delta r|. \quad (10)$$

At this root, the N th step lands on the starting roller after exactly one traversal of the locus. Equation (1) is therefore satisfied for all links, including the closing link.

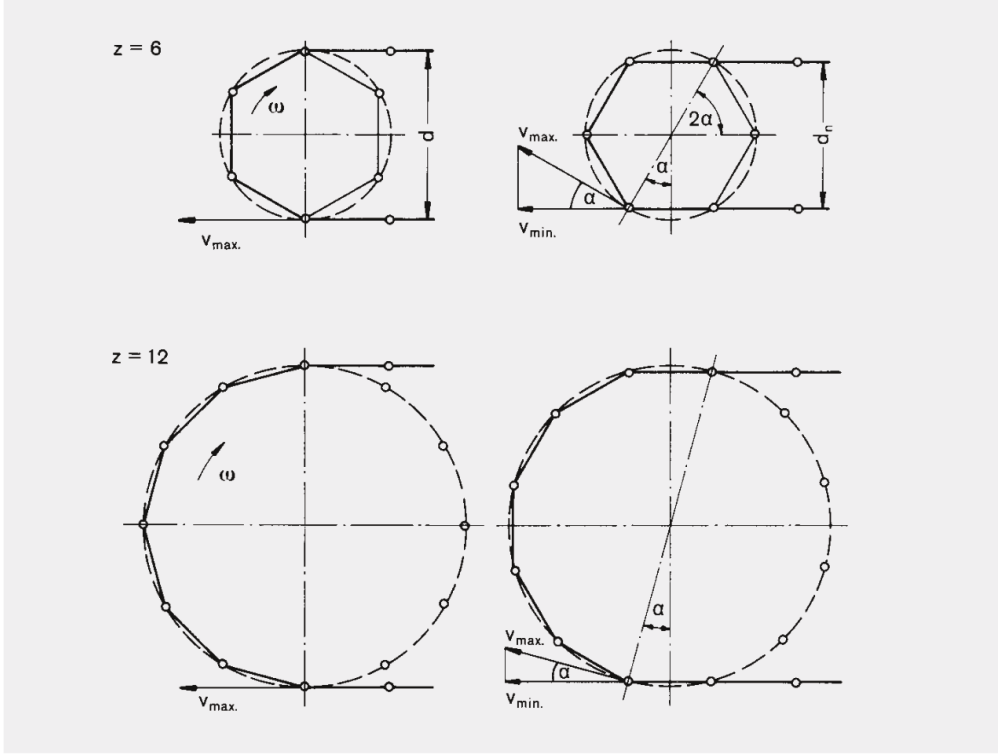


Figure 2: The pitch-circle curve is continuous, but the physical chain is a sequence of rigid pitch chords between roller centers.

4 Integer-link selection

The algorithm evaluates candidate counts in a configurable neighborhood of N_0 . If the even-only option is active, odd candidates are removed. For every remaining N , Equation (10) is solved by an outer bracketing and bisection procedure. The selected candidate minimizes

$$|a_f(N) - a_0|. \quad (11)$$

This selection rule makes the requested center distance the design target while maintaining exact discrete closure. The result also records

$$\Delta a = a_f - a_0, \quad (12)$$

$$C_d = Np, \quad (13)$$

$$e_{\max} = \max_i ||P_{i+1} - P_i|| - p. \quad (14)$$

5 ASA 80 regression example

The reference case uses ASA 80-1 chain with pitch $p = 25.400$ mm, $z_1 = 11$, $z_2 = 20$, and $a_0 = 400.000$ mm. The pitch diameters are

$$D_{p1} = 90.156\,424\,5 \text{ mm}, \quad D_{p2} = 162.368\,311\,8 \text{ mm}. \quad (15)$$

At the desired center distance, the continuous locus has length

$$C_i(a_0) = 1199.926\,244\,5 \text{ mm}, \quad \frac{C_i(a_0)}{p} = 47.2411907. \quad (16)$$

When odd link counts are allowed, the nearest exact rigid-link solution is summarized in Table 1.

Table 1: Nearest rigid-link solution for the ASA 80 regression case.

Quantity	Value
Link count, N	47
Corrected center distance, a_f	398.338 658 801 mm
Center correction, Δa	-1.661 341 199 mm
Discrete chain length, Np	1193.800 000 000 mm
Continuous locus at a_f	1196.617 182 662 mm
Closure residual, $g(a_f; N)$	2.110×10^{-10} mm
Maximum pitch error, $e_{\max}$	2.104×10^{-10} mm
Offset link required	Yes

The nonzero difference between the continuous locus length and Np is expected: circular arcs are longer than their inscribed pitch chords. What must converge to zero is the discrete residual in Equation (9), together with the chord error in Equation (1).

If even counts are required, the selected solution is $N = 48$ with $a_f = 411.109\,006\,689$ mm, closure residual -9.845×10^{-11} mm, maximum pitch error 9.902×10^{-11} mm, and no offset link.

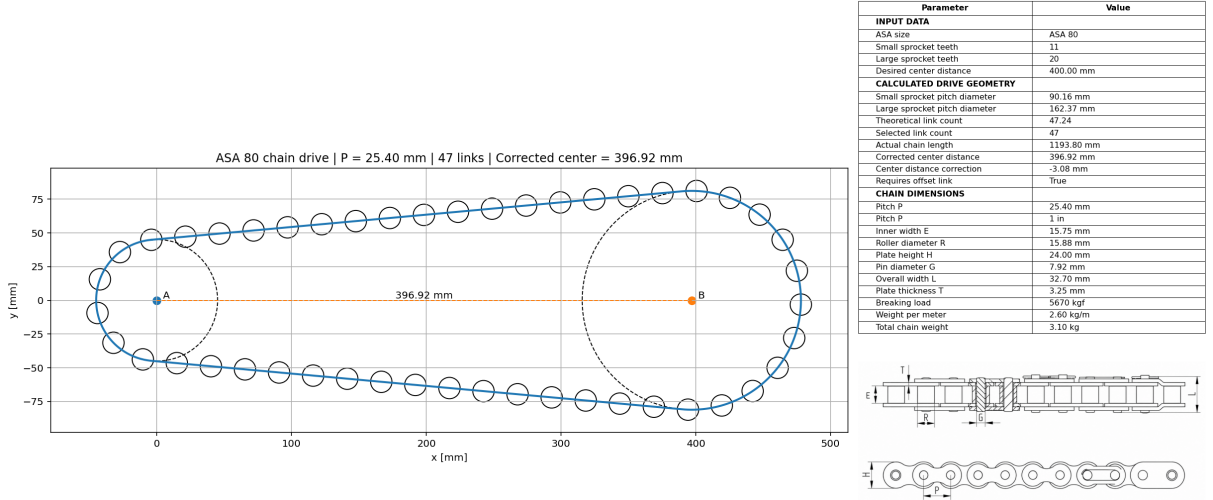


Figure 3: Validated ASA 80 result. Blue denotes the continuous roller-center locus; orange segments are the rigid pitch chords.

6 Computational implementation and validation

The dependency-free solver returns the roller centers and one pose per link: position, in-plane angle, pitch length and type (inner, outer or offset). A compatibility layer adds catalog data, pitch dimensions, report groups and chain-mass estimation for the command-line and Streamlit interfaces.

Three automated regression tests are included:

1. ASA 80, 11T/20T, 400 mm, odd counts allowed;
2. the same case with the even-link constraint;
3. ASA 80, 9T/10T, desired center $6.5p$, which selects 23 links and one offset link.

For every case the tests independently recompute every consecutive Euclidean distance, verify the link count and parity, and require both the closure residual and maximum pitch error to remain below 1×10^{-8} mm.

7 Scope and limitations

The formulation covers nominal geometric closure for two pitch-perfect, coplanar sprockets and a new chain. It does not evaluate transmitted power, service factors, tension, fatigue, lubrication, wear, chain elongation, manufacturer-specific ratings, sprocket tooth strength, shaft loads, bearing loads or dynamic polygonal action. An offset link may carry a lower allowable load than a standard alternating assembly; the applicable chain manufacturer's guidance must be followed.

8 Conclusion

The continuous pitch-circle path remains valuable as a locus and as an initial estimate, but equating its arc length to Np is not a physically exact closure condition for roller chain. Walking rigid pitch chords and solving the discrete residual produces roller positions that can be transferred directly to CAD and validated link by link. For the ASA 80 regression case, the corrected center is 398.338 658 801 mm for 47 links, rather than the superseded continuous- length result of approximately 396.92 mm.

References

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